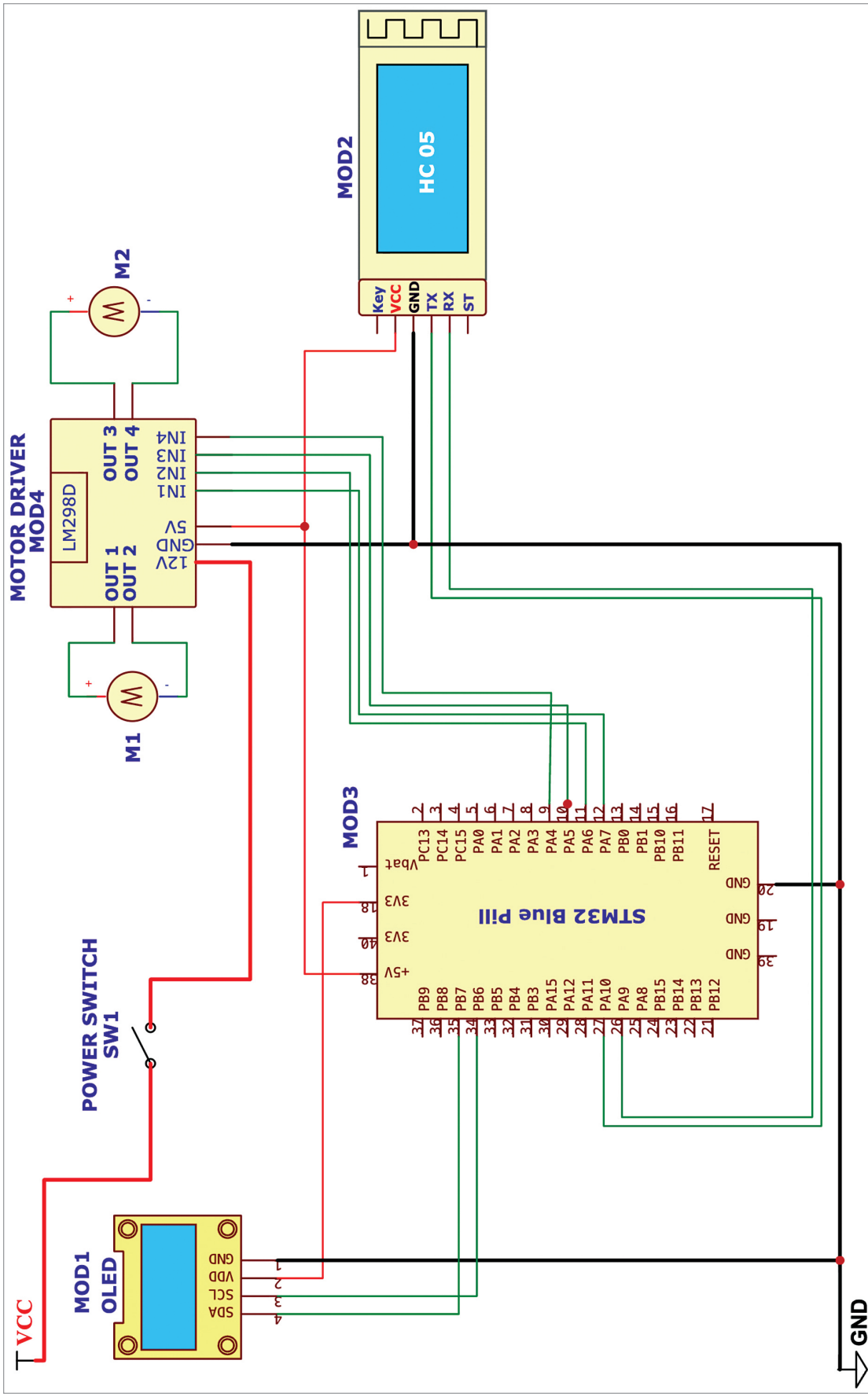



*Fig. 3: Circuit diagram*

2 shows the wiring diagram of the system. The components required to build the system are listed in the Bill of Materials table.

## Circuit and working

Fig. 3 shows the circuit diagram of the Bluetooth-controlled robot using the STM32 Blue Pill. The system is built around an STM32F103C8T6-based Blue Pill board, an L298N motor driver, an HC-05 Bluetooth module, four dual-shaft 200rpm BO geared motors, a 0.96-inch (2.4cm) I2C OLED display, two 3.7V, 2000mAh-2500mAh Li-ion batteries, a two-cell holder, and an SPST switch.

The robot is designed as a compact, four-wheel mobile platform with a 3D-printed chassis that provides stability and balanced load distribution. Four BO geared motors are mounted on the wheels to provide sufficient torque. The STM32F103C8T6 operates at 72MHz and manages Bluetooth commands,